Quantification: 535_z27c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	31
interior lacuna count	27
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 27)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2829.91	1681.5	2856.35	px^2
major_axis_length_px	lacuna shape	83.8653	65.9023	42.0739	px
minor_axis_length_px	lacuna shape	42.4325	36.574	22.2519	px
aspect_ratio	lacuna shape	2.0977	2.1684	0.7308	unitless
eccentricity	lacuna shape	0.822	0.8873	0.1326	unitless
solidity	lacuna shape	0.7916	0.7951	0.0919	unitless
roots_count	headline	6.2593	5	3.8189	count
ring_length_r30_px	headline	203.185	180	103.775	px
ring_length_r60_px	reported	431.741	363	237.505	px
owned_length_px	ownership-dependent	452.936	251.309	546.801	px
edge_count	ownership-dependent	21.1481	11	22.6863	count
mean_edge_length_px	ownership-dependent	19.8495	19.5648	7.5656	px
perimeter_px	normalised	234.126	184.853	157.984	px
ring_area_r30_px2	normalised	7682.59	6087	4098.39	px^2
ring_area_r60_px2	normalised	16338.1	14824	8404.35	px^2
in_frame_fraction_r30	normalised	0.9821	1	0.0715	unitless
in_frame_fraction_r60	normalised	0.9666	1	0.1125	unitless
ring_density_r30	normalised	0.0280344	0.0266033	0.00945202	px^-1
ring_density_r60	normalised	0.0271317	0.0276445	0.0074474	px^-1
roots_per_100px_perimeter	normalised	3.1138	2.5142	1.5458	per 100 px
ring_attached_length_r30_px	network v2	165.741	139	101.938	px
ring_attached_length_r60_px	network v2	271.778	217	217.007	px
ring_length_w_r30_px	network v2	229.576	199.61	117.068	px
ring_length_w_r60_px	network v2	487.552	409.777	271.91	px
sholl_crossings_r10	network v2	5.2593	4	3.2769	count
sholl_crossings_r20	network v2	6.037	5	3.8178	count
sholl_crossings_r30	network v2	6.1852	5	3.9129	count
loop_count	loops, report only	0.7778	0	1.8674	count
loop_edge_share	loops, report only	0.0413	0	0.0885	share
ring_width_mean_r30_px	width	7.7039	7.8006	2.0355	px

Per field
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measure	value	units
analysed_area_px2	970960	px^2
total_skeleton_length_px	24026	px
canalicular_length_density_per_px	0.0247446	px^-1
junction_count	433	count
junction_density_per_px2	0.00044595	px^-2
lacuna_count	31	count

lacunae_per_px2	3.193e-05	px^-2
skeleton_component_count	413	count
mean_component_length_px	58.1743	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0278984	px^-1
field_density_without_flagged_per_px	0.0232787	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1258	px
unattached_component_count	76	count
canalicular_length_density_attached_per_px	0.0247446	px^-1
field_length_density_with_unattached_per_px	0.0260402	px^-1
loop_count	30	count
loop_edge_share	0.0863	share
edge_count	1263	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5683	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	18822	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	414	38.6201	15.6281	2.4712	0.9145	0.7709	-0.8461	49.0749	570.635	False
2	2942	101.888	45.2188	2.2532	0.8961	0.7542	-1.1126	105.425	685.744	False
3	810	40.5726	31.6629	1.2814	0.6253	0.7226	0.1823	107.814	615.768	False
4	957	54.4409	24.7864	2.1964	0.8903	0.7988	0.0308	119.021	790.769	False
5	1484	55.1992	46.0237	1.1994	0.5521	0.6199	-0.2191	163.815	808.798	False
6	955	64.3571	19.6453	3.276	0.9523	0.8459	1.0779	145.542	454.49	False
7	3599	102.364	47.1738	2.1699	0.8875	0.8955	-1.0474	233.848	107.369	False
8	1807	63.8235	38.7415	1.6474	0.7947	0.9372	1.5189	218.143	826.646	False
9	712	35.4872	28.8379	1.2306	0.5828	0.9368	-0.3495	301.131	1012.33	True
10	11051	179.07	107.127	1.6716	0.8013	0.7079	0.7029	439.405	253.828	False
11	5312	120.404	59.815	2.0129	0.8679	0.8459	-1.268	460.68	557.419	False
12	1257	48.9366	34.4065	1.4223	0.7111	0.929	1.2464	562.555	134.776	False
13	796	41.9622	30.4426	1.3784	0.6882	0.761	0.6839	561.568	949.678	False
14	2123	56.2494	49.4219	1.1381	0.4775	0.9222	0.3101	574.703	493.107	False
15	2990	94.2448	43.1713	2.183	0.8889	0.8771	0.9473	600.532	860.511	False
16	1556	70.3776	32.2729	2.1807	0.8887	0.7988	1.1709	626.462	984.626	False
17	1310	65.9258	27.032	2.4388	0.9121	0.8501	-1.1711	624.014	302.034	False
18	9402	173.447	80.0495	2.1668	0.8871	0.7022	-0.8455	691.761	675.898	False
19	2231	116.941	28.0885	4.1633	0.9707	0.7914	0.9981	758.602	307.963	False
20	865	65.8788	20.5864	3.2001	0.9499	0.7509	1.3168	799.104	703.183	False
21	6401	142.513	65.0238	2.1917	0.8898	0.8153	-1.2115	832.616	623.668	False
22	3509	107.726	65.7294	1.6389	0.7923	0.6187	-0.7908	851.443	291.978	False
23	1656	75.1349	31.8764	2.3571	0.9055	0.792	-0.7612	861.306	996.914	True
24	487	40.0944	21.468	1.8676	0.8446	0.6987	-0.7102	944.786	749.624	False

25	754	37.3455	26.9997	1.3832	0.6909	0.9286	-0.1889	961.662	1011.9	True
26	10983	260.5	56.6496	4.5984	0.9761	0.8482	-1.5452	998.074	728.234	True
27								17.7078	1000.47	False
28								225.615	527.407	False
29								327.698	939.349	False
30								535.548	870.784	False
31								883.569	853.096	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	282	463	76.3553	7	10.9079	8.0265
2	6	280	535	792.097	46	17.2195	9.5449
3	8	197	362	231.652	17	13.6266	2.8284
4	4	133	236	136.953	7	19.5648	8.3237
5	7	208	337	255.924	11	23.2658	8.0191
6	3	109	327	75.4975	5	15.0995	
7	6	284	834	1790.48	84	21.3152	7.1459
8	4	140	363	101.154	4	25.2886	7.8006
9	2	87	168	98.7696	5	19.7539	7.5705
10	15	445	1043	1159.97	51	22.7446	7.7299
11	18	336	754	935.754	48	19.4949	8.3125
12	7	161	457	525.002	17	30.8825	8.5498
13	3	76	118	37.2132	5	7.4426	3.9876
14	5	180	387	295.794	9	32.866	8.4103
15	9	215	365	251.309	11	22.8462	6.4255
16	6	148	290	300.836	12	25.0696	9.9005
17	3	142	269	147.154	4	36.7886	7.8455
18	12	462	975	1978.3	77	25.6922	7.637
19	4	190	407	331.35	13	25.4885	7.2625
20	4	135	257	332.137	24	13.839	7.0919
21	7	181	432	319.664	15	21.3109	7.2398
22	10	385	744	1498.33	53	28.2704	7.1584
23	4	121	206	126.154	6	21.0257	7.0065
24	6	131	268	118.598	11	10.7816	5.6728
25	2	77	153	125.811	5	25.1622	8.2557
26	11	317	692	960.08	64	15.0013	7.1245
27	4	126	238	84.0833	8	10.5104	5.7183
28	3	94	290	95.9828	8	11.9978	
29	5	197	415	118.083	9	13.1204	10.5676
30	4	101	151	169.723	9	18.8581	13.3629
31	1	148	340	69.8701	6	11.645	8.0361

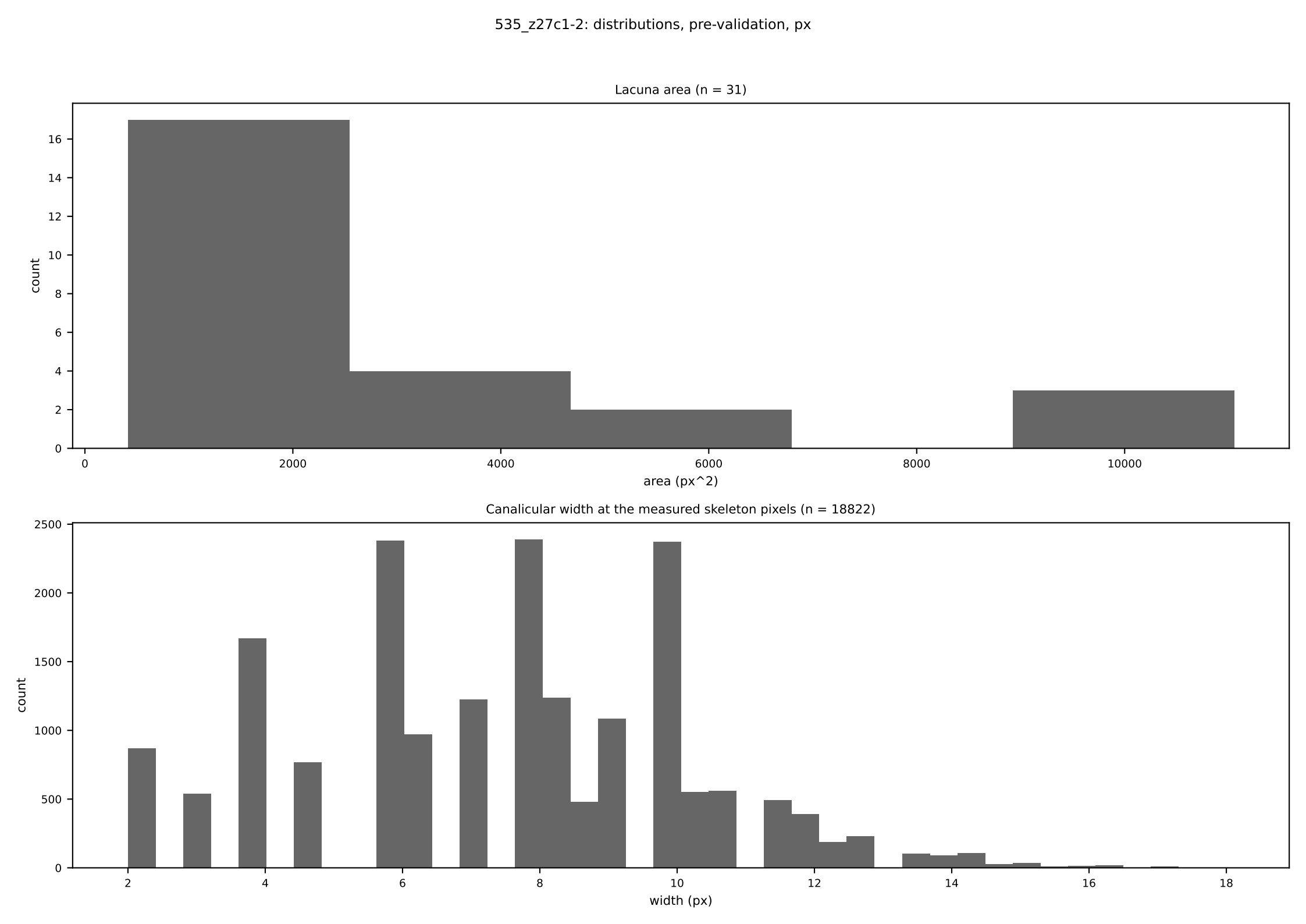
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	117.64	5345	11307	1	0.893	0.0527596	0.0409481	4.2503
2	369.203	9004	16769	1	1	0.0310973	0.0319041	1.6251
3	188.102	4925	8909	1	1	0.04	0.0406331	4.253
4	159.095	4461	7968	1	1	0.0298139	0.0296185	2.5142
5	254.072	5329	9525	1	1	0.0390317	0.0353806	2.7551
6	163.723	7120	17892	1	1	0.015309	0.0182763	1.8324
7	281.948	10538	26647	1	1	0.0269501	0.0312981	2.128
8	184.853	6087	14824	1	1	0.0229998	0.0244873	2.1639
9	112.426	3391	6785	0.5858	0.5585	0.0256561	0.0247605	1.7789
10	645.151	19717	40885	1	1	0.0225694	0.0255106	2.325
11	368.818	12630	27408	1	1	0.0266033	0.0275102	4.8805

12	147.054	6972	18793	1	1	0.0230924	0.0243176	4.7602
13	153.338	5222	9351	1	0.9843	0.0145538	0.012619	1.9565
14	184.267	7991	18400	1	1	0.0225253	0.0210326	2.7135
15	250.392	8593	16094	1	1	0.0250204	0.0226793	3.5944
16	219.35	5949	9604	0.8676	0.7611	0.0248781	0.0301958	2.7354
17	195.48	7849	18629	1	1	0.0180915	0.0144399	1.5347
18	634.625	17290	32472	1	1	0.0267206	0.0300259	1.8909
19	284.534	9498	19681	1	1	0.0200042	0.0206798	1.4058
20	164.646	4538	8768	1	1	0.0297488	0.0293111	2.4295
21	465.843	11554	22109	1	1	0.0156656	0.0195395	1.5027
22	402.073	11840	23982	1	1	0.0325169	0.0310233	2.4871
23	212.551	6244	12720	0.731	0.6597	0.0193786	0.016195	1.8819
24	129.196	3723	7613	1	0.9986	0.0351867	0.0352029	4.6441
25	116.805	3670	7367	0.6042	0.5352	0.0209809	0.0207683	1.7123
26	816.873	11314	20728	0.621	0.5937	0.0280184	0.0333848	1.3466
27	57.9411	2894	6717	0.648	0.4609	0.0435384	0.0354325	6.9036
28	84.669	5109	13944	1	1	0.0183989	0.0207975	3.5432
29	73.0122	4735	12338	1	1	0.0416051	0.0336359	6.8482
30	87.4975	4160	8201	1	1	0.0242789	0.0184124	4.5716
31	54.8701	4357	12299	1	1	0.0339683	0.0276445	1.8225

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	194	281	315.279	517.718	7	11	9
2	254	405	319.664	604.529	8	10	8
3	152	206	222.823	409.777	4	7	6
4	122	122	153.368	267.38	5	4	2
5	162	228	230.095	376.978	4	7	5
6	57	65	126.61	376.647	2	5	7
7	224	543	320.108	931.725	6	8	11
8	90	99	161.853	406.576	3	4	6
9	80	84	93.9411	188.61	3	3	2
10	381	718	507.902	1193.57	14	15	15
11	298	631	369.907	837.683	9	10	9
12	157	331	186.823	522.002	5	5	6
13	24	24	89.8112	131.711	1	2	0
14	139	231	199.61	432.433	4	5	8
15	145	217	244.622	411.262	6	5	4
16	97	162	169.61	327.25	3	5	3
17	107	184	157.297	299.279	4	5	4
18	457	859	526.931	1136.06	14	16	15
19	169	254	210.539	456.848	5	5	6
20	135	241	147.598	289.096	5	3	3
21	168	230	204.51	487.647	9	2	5
22	342	605	434.877	842.826	7	10	14
23	84	118	133.811	231.238	5	4	4
24	123	129	145.326	293.551	3	1	3
25	68	100	87.0122	169.953	2	2	3
26	283	623	350.25	779.683	8	7	8
27	88	88	145.782	272.664	3	4	4
28	79	83	109.054	326.806	2	2	4
29	90	106	212.782	456.233	2	5	4
30	87	87	121.368	177.238	3	2	2
31	134	209	164.397	378.421	4	5	4


535_z27c1-2: distributions, pre-validation, px
Lacuna area (n = 31)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 18822)
count
width (px)
0
500
1000
1500
2000
2500
2
4
6
8
10
12
14
16
18